

# Multiple Regression

MGMT 670: Business Analytics



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## Regression Models

- Describe the relationship between a response (or dependent) variable and one or more explanatory (or independent) variables.
- Used for prediction
- Simple vs. Multiple Regression

## Outline

- Adjusted R-square
- Different tests
  - Overall test;
  - Individual test;
  - partial F-test
- Multicollinearity
  - VIF
- Dummy variable

## Multiple Linear Regression Model

- Relationship between 1 dependent variable and 1 or more independent variables is a linear function.

The diagram shows the equation  $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$  with arrows pointing from descriptive labels to each part of the equation:

- Population Y-intercept** points to  $\beta_0$
- Population slopes** points to the coefficients  $\beta_1, \beta_2, \dots, \beta_k$
- Random error** points to  $\varepsilon$
- Dependent (response) variable** points to  $Y$
- Independent (explanatory) variables** points to the  $X$  variables ( $X_1, X_2, \dots, X_k$ )

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# Estimated Multiple Regression Equation

- Multiple Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

$$\varepsilon \sim N(0, \sigma)$$

- Estimated Multiple Regression Equation  
(Prediction equation)

$$\hat{Y} = b_0 + b_1 X_1 + b_2 X_2 + \dots + b_k X_k$$

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# Estimate Parameters

## Least Squares Method

Find the estimates for  $\beta_0, \beta_1, \dots, \beta_k$ , which Minimize the sum of squared errors:

$$\sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n e_i^2$$

Computation of coefficients' values involves matrix algebra. We will rely on computer software packages.

$$\text{estimate of } \sigma \text{ is } s = \sqrt{\frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{n - k - 1}} = \sqrt{\frac{\sum_{i=1}^n e_i^2}{n - k - 1}}$$

$n - k - 1$  is degree of freedom for unexplained variations

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## Who is your competitor?



- You are a new marketing analyst at Hannah Solar, a manufacturer that produces solar panels to the retail market. This morning, you received a memo indicating that **Efacec USA, a competing manufacturer, may substantially lower the price of their products.** Your company is concerned about the negative impact this might have on its market share.
- Andrew, one of your colleagues, thinks that **Petra Solar is Hannah Solar's main competitor and that Efacec's new campaign will NOT substantially influence Hannah Solar's market share.** This is because, Andrew argues, Efacec's solar panels are crystalline silicon-based, while both Hannah and Petra's products are made with thin film technology, which is cheaper but has lower level of energy efficiency. Furthermore, Efacec has two different products, type I and type II, while both Petra and your company, Hannah, has only one major product.

# Solar panels production

- Your company: Hannah Solar
  - Produces one type of solar panels
- Petra Solar
  - Produces one and the same type of solar panels as Hannah
- Efacec USA
  - Produces two different types of solar panels

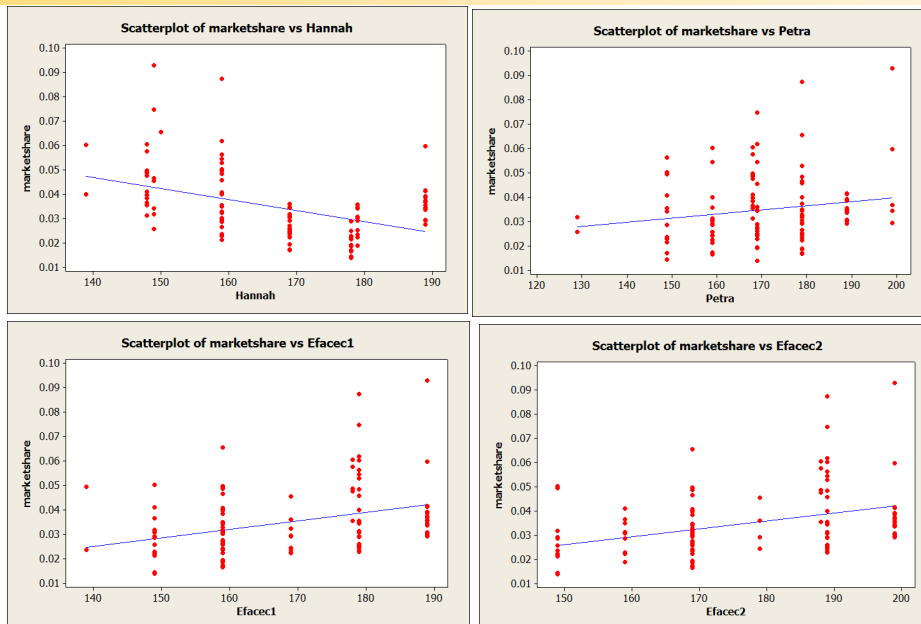


Efacec USA is changing their prices,

How does it impact your company market share?

# Data n=113

| week | marketshare | Hannah | Petra | Efacec1 | Efacec2 |
|------|-------------|--------|-------|---------|---------|
| 1    | 0.0454565   | 149    | 169   | 169     | 179     |
| 2    | 0.0930145   | 149    | 199   | 189     | 199     |
| 3    | 0.0596656   | 189    | 199   | 189     | 199     |
| 4    | 0.0345966   | 189    | 199   | 189     | 199     |
| 5    | 0.0276536   | 189    | 169   | 159     | 169     |
| 6    | 0.0294224   | 189    | 199   | 189     | 199     |
| 7    | 0.0368832   | 189    | 199   | 189     | 199     |
| 8    | 0.0413369   | 189    | 189   | 189     | 199     |
| 9    | 0.0291296   | 189    | 189   | 169     | 179     |
| 10   | 0.0391639   | 189    | 189   | 189     | 199     |
| 11   | 0.0335741   | 189    | 189   | 189     | 199     |
| 12   | 0.0345275   | 189    | 179   | 189     | 199     |
| 13   | 0.0415441   | 189    | 189   | 189     | 199     |
| 14   | 0.0354724   | 189    | 189   | 189     | 199     |
| 15   | 0.0372911   | 189    | 179   | 189     | 199     |
| 16   | 0.0387742   | 189    | 189   | 189     | 199     |
| 17   | 0.0346312   | 179    | 189   | 179     | 189     |
| 18   | 0.0308691   | 179    | 189   | 189     | 199     |
| 19   | 0.0306231   | 179    | 189   | 189     | 199     |
| ...  |             |        |       |         |         |
| 113  |             |        |       |         |         |



## Regression Analysis: marketshare versus Hannah, Petra, Efacec1, Efacec2

### Analysis of Variance

| Source      | DF  | Adj SS   | Adj MS   | F-Value | P-Value |
|-------------|-----|----------|----------|---------|---------|
| Regression  | 4   | 0.012014 | 0.003003 | 30.00   | 0.000   |
| Hannah      | 1   | 0.008826 | 0.008826 | 88.16   | 0.000   |
| Petra       | 1   | 0.000969 | 0.000969 | 9.68    | 0.002   |
| Efacec1     | 1   | 0.000110 | 0.000110 | 1.10    | 0.297   |
| Efacec2     | 1   | 0.000012 | 0.000012 | 0.12    | 0.728   |
| Error       | 108 | 0.010812 | 0.000100 |         |         |
| Lack-of-Fit | 49  | 0.006692 | 0.000137 | 1.96    | 0.007   |
| Pure Error  | 59  | 0.004120 | 0.000070 |         |         |
| Total       | 112 | 0.022826 |          |         |         |

### Model Summary

| S         | R-sq   | R-sq(adj) | R-sq(pred) |
|-----------|--------|-----------|------------|
| 0.0100055 | 52.63% | 50.88%    | 47.57%     |

### Coefficients

| Term     | Coef      | SE Coef  | T-Value | P-Value | VIF   |
|----------|-----------|----------|---------|---------|-------|
| Constant | 0.0403    | 0.0141   | 2.85    | 0.005   |       |
| Hannah   | -0.000760 | 0.000081 | -9.39   | 0.000   | 1.36  |
| Petra    | 0.000262  | 0.000084 | 3.11    | 0.002   | 1.66  |
| Efacec1  | 0.000347  | 0.000332 | 1.05    | 0.297   | 25.97 |
| Efacec2  | 0.000102  | 0.000294 | 0.35    | 0.728   | 25.15 |

### Regression Equation

$$\text{marketshare} = 0.0403 - 0.000760 \text{ Hannah} + 0.000262 \text{ Petra} + 0.000347 \text{ Efacec1} + 0.000102 \text{ Efacec2}$$

## Coefficient of Multiple Determination

- Proportion of Y variation “explained” by all X variables.

$$R^2 = \frac{\text{Explained Variation}}{\text{Total Variation}} = \frac{SSR}{SST}$$

- Non-decreasing** with number of variables



**Should we keep adding variables?**

## Adjusted $R^2$

- Adjust  $R^2$  to account for number of variables
- Reflects
  - Sample size
  - Number of independent variables (k)

$$R_a^2 = 1 - \frac{SSE / (n - k - 1)}{SST / (n - 1)}$$

- Adjust  $R^2$  is always smaller than  $R^2$**
- Used to compare models with different # of variables

### Regression Analysis: marketshare versus Hannah, Petra, Efacec1, Efacec2

#### Analysis of Variance

| Source      | DF  | Adj SS   | Adj MS   | F-Value | P-Value |
|-------------|-----|----------|----------|---------|---------|
| Regression  | 4   | 0.012014 | 0.003003 | 30.00   | 0.000   |
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#### Regression Equation

marketshare = 0.0403 - 0.000760 Hannah + 0.000262 Petra + 0.000347 Efacec1 + 0.000102 Efacec2

$$\text{Adjusted } R^2 = 1 - \frac{SSE / (n - k - 1)}{SST / (n - 1)} =$$

## Testing significance

- Overall test: is there a significant relationship between all prices and your market share?
- Individual test: is there a significant relationship between a single and your market share?
- Joint test: is there a significant relationship between a group of prices and your market share?
- These are all the same in simple regression**

## Testing Overall Significance

- Shows if there is a linear relationship between all X variables together and Y
- Uses F test statistic
- Hypotheses
  - $H_0: \beta_1 = \beta_2 = \dots = \beta_k = 0$  (No linear relationship)
  - $H_A$ : At least one  $\beta_i \neq 0$  (Linear relationship; At least one X variable affects Y.)

### Regression Analysis: marketshare versus Hannah, Petra, Efacec1, Efacec2

#### Analysis of Variance

| Source      | DF  | Adj SS   | Adj MS   | F-Value | P-Value |
|-------------|-----|----------|----------|---------|---------|
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$$F = \frac{MSR}{MSE} = \frac{0.0030035}{0.0001001} = 30.00$$

Critical F value ( $\alpha = 0.05$ )

$$F_{\alpha, k, n-k-1} = 2.45$$

**Conclusion:** F-test > critical value

there is a significant relationship between Hannah's market share and all prices.

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### Regression Analysis: marketshare versus Hannah, Petra, Efacec1, Efacec2

#### Analysis of Variance

| Source      | DF  | Adj SS   | Adj MS   | F-Value | P-Value |
|-------------|-----|----------|----------|---------|---------|
| Regression  | 4   | 0.012014 | 0.003003 | 30.00   | 0.000   |
| Hannah      | 1   | 0.008826 | 0.008826 | 88.16   | 0.000   |
| Petra       | 1   | 0.000969 | 0.000969 | 9.68    | 0.002   |
| Efacec1     | 1   | 0.000110 | 0.000110 | 1.10    | 0.297   |
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#### Model Summary

| S         | R-sq   | R-sq(adj) | R-sq(pred) |
|-----------|--------|-----------|------------|
| 0.0100055 | 52.63% | 50.88%    | 47.57%     |

#### Coefficients

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| Hannah   | -0.000760 | 0.000081 | -9.39   | 0.000   | 1.36  |
| Petra    | 0.000262  | 0.000084 | 3.11    | 0.002   | 1.66  |
| Efacec1  | 0.000347  | 0.000332 | 1.05    | 0.297   | 25.97 |
| Efacec2  | 0.000102  | 0.000294 | 0.35    | 0.728   | 25.15 |

P-value=

**Conclusion:** p-value <  $\alpha$

there is a significant relationship between Hannah's market share and all prices.

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## Testing Individual Significance

- Does a given independent variable have a significant relation with response? (in the presence of other independent variables)
- Uses t test statistic
- Hypothesis
  - $H_0: \beta_i = 0$  (No linear relationship)
  - $H_A: \beta_i \neq 0$  (Linear relationship with i-th independent variable)

### Regression Analysis: marketshare versus Hannah, Petra, Efacec1, Efacec2

#### Analysis of Variance

| Source      | DF  | Adj SS   | Adj MS   | F-Value | P-Value |
|-------------|-----|----------|----------|---------|---------|
| Regression  | 4   | 0.012014 | 0.003003 | 30.00   | 0.000   |
| Hannah      | 1   | 0.008826 | 0.008826 | 88.16   | 0.000   |
| Petra       | 1   | 0.000969 | 0.000969 | 9.68    | 0.002   |
| Efacec1     | 1   | 0.000110 | 0.000110 | 1.10    | 0.297   |
| Efacec2     | 1   | 0.000012 | 0.000012 | 0.12    | 0.728   |
| Error       | 108 | 0.010812 | 0.000100 |         |         |
| Lack-of-Fit | 49  | 0.006692 | 0.000137 | 1.96    | 0.007   |
| Pure Error  | 59  | 0.004120 | 0.000070 |         |         |
| Total       | 112 | 0.022826 |          |         |         |

#### Model Summary

| S         | R-sq   | R-sq(adj) | R-sq(pred) |
|-----------|--------|-----------|------------|
| 0.0100055 | 52.63% | 50.88%    | 47.57%     |

#### Coefficients

| Term     | Coef      | SE Coef  | T-Value | P-Value | VIF   |
|----------|-----------|----------|---------|---------|-------|
| Constant | 0.0403    | 0.0141   | 2.85    | 0.005   |       |
| Hannah   | -0.000760 | 0.000081 | -9.39   | 0.000   | 1.36  |
| Petra    | 0.000262  | 0.000084 | 3.11    | 0.002   | 1.66  |
| Efacec1  | 0.000347  | 0.000332 | 1.05    | 0.297   | 25.97 |
| Efacec2  | 0.000102  | 0.000294 | 0.35    | 0.728   | 25.15 |

Is Hannah Significant with 5% level?

**Conclusion:  $p\text{-value} < \alpha$**

there is a significant relationship between Hannah's market share and Hannah price

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### Regression Analysis: marketshare versus Hannah, Petra, Efacec1, Efacec2

#### Analysis of Variance

| Source      | DF  | Adj SS   | Adj MS   | F-Value | P-Value |
|-------------|-----|----------|----------|---------|---------|
| Regression  | 4   | 0.012014 | 0.003003 | 30.00   | 0.000   |
| Hannah      | 1   | 0.008826 | 0.008826 | 88.16   | 0.000   |
| Petra       | 1   | 0.000969 | 0.000969 | 9.68    | 0.002   |
| Efacec1     | 1   | 0.000110 | 0.000110 | 1.10    | 0.297   |
| Efacec2     | 1   | 0.000012 | 0.000012 | 0.12    | 0.728   |
| Error       | 108 | 0.010812 | 0.000100 |         |         |
| Lack-of-Fit | 49  | 0.006692 | 0.000137 | 1.96    | 0.007   |
| Pure Error  | 59  | 0.004120 | 0.000070 |         |         |
| Total       | 112 | 0.022826 |          |         |         |

#### Model Summary

| S         | R-sq   | R-sq(adj) | R-sq(pred) |
|-----------|--------|-----------|------------|
| 0.0100055 | 52.63% | 50.88%    | 47.57%     |

#### Coefficients

| Term     | Coef      | SE Coef  | T-Value | P-Value | VIF   |
|----------|-----------|----------|---------|---------|-------|
| Constant | 0.0403    | 0.0141   | 2.85    | 0.005   |       |
| Hannah   | -0.000760 | 0.000081 | -9.39   | 0.000   | 1.36  |
| Petra    | 0.000262  | 0.000084 | 3.11    | 0.002   | 1.66  |
| Efacec1  | 0.000347  | 0.000332 | 1.05    | 0.297   | 25.97 |
| Efacec2  | 0.000102  | 0.000294 | 0.35    | 0.728   | 25.15 |

Is Petra significant with 5% level?

#### Regression Equation

marketshare = 0.0403 - 0.000760 Hannah + 0.000262 Petra + 0.000347 Efacec1 + 0.000102 Efacec2

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### Regression Analysis: marketshare versus Hannah, Petra, Efacec1, Efacec2

#### Analysis of Variance

| Source      | DF  | Adj SS   | Adj MS   | F-Value | P-Value |
|-------------|-----|----------|----------|---------|---------|
| Regression  | 4   | 0.012014 | 0.003003 | 30.00   | 0.000   |
| Hannah      | 1   | 0.008826 | 0.008826 | 88.16   | 0.000   |
| Petra       | 1   | 0.000969 | 0.000969 | 9.68    | 0.002   |
| Efacec1     | 1   | 0.000110 | 0.000110 | 1.10    | 0.297   |
| Efacec2     | 1   | 0.000012 | 0.000012 | 0.12    | 0.728   |
| Error       | 108 | 0.010812 | 0.000100 |         |         |
| Lack-of-Fit | 49  | 0.006692 | 0.000137 | 1.96    | 0.007   |
| Pure Error  | 59  | 0.004120 | 0.000070 |         |         |
| Total       | 112 | 0.022826 |          |         |         |

#### Model Summary

| S         | R-sq   | R-sq(adj) | R-sq(pred) |
|-----------|--------|-----------|------------|
| 0.0100055 | 52.63% | 50.88%    | 47.57%     |

#### Coefficients

| Term     | Coef      | SE Coef  | T-Value | P-Value | VIF   |
|----------|-----------|----------|---------|---------|-------|
| Constant | 0.0403    | 0.0141   | 2.85    | 0.005   |       |
| Hannah   | -0.000760 | 0.000081 | -9.39   | 0.000   | 1.36  |
| Petra    | 0.000262  | 0.000084 | 3.11    | 0.002   | 1.66  |
| Efacec1  | 0.000347  | 0.000332 | 1.05    | 0.297   | 25.97 |
| Efacec2  | 0.000102  | 0.000294 | 0.35    | 0.728   | 25.15 |

Is Efacec1 significant with 5% level?

#### Regression Equation

marketshare = 0.0403 - 0.000760 Hannah + 0.000262 Petra + 0.000347 Efacec1 + 0.000102 Efacec2

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### Regression Analysis: marketshare versus Hannah, Petra, Efacec1, Efacec2

#### Analysis of Variance

| Source      | DF  | Adj SS   | Adj MS   | F-Value | P-Value |
|-------------|-----|----------|----------|---------|---------|
| Regression  | 4   | 0.012014 | 0.003003 | 30.00   | 0.000   |
| Hannah      | 1   | 0.008826 | 0.008826 | 88.16   | 0.000   |
| Petra       | 1   | 0.000969 | 0.000969 | 9.68    | 0.002   |
| Efacec1     | 1   | 0.000110 | 0.000110 | 1.10    | 0.297   |
| Efacec2     | 1   | 0.000012 | 0.000012 | 0.12    | 0.728   |
| Error       | 108 | 0.010812 | 0.000100 |         |         |
| Lack-of-Fit | 49  | 0.006692 | 0.000137 | 1.96    | 0.007   |
| Pure Error  | 59  | 0.004120 | 0.000070 |         |         |
| Total       | 112 | 0.022826 |          |         |         |

#### Model Summary

| S         | R-sq   | R-sq(adj) | R-sq(pred) |
|-----------|--------|-----------|------------|
| 0.0100055 | 52.63% | 50.88%    | 47.57%     |

#### Coefficients

| Term     | Coef      | SE Coef  | T-Value | P-Value | VIF   |
|----------|-----------|----------|---------|---------|-------|
| Constant | 0.0403    | 0.0141   | 2.85    | 0.005   |       |
| Hannah   | -0.000760 | 0.000081 | -9.39   | 0.000   | 1.36  |
| Petra    | 0.000262  | 0.000084 | 3.11    | 0.002   | 1.66  |
| Efacec1  | 0.000347  | 0.000332 | 1.05    | 0.297   | 25.97 |
| Efacec2  | 0.000102  | 0.000294 | 0.35    | 0.728   | 25.15 |

Is Efacec2 significant with 5% level?

#### Regression Equation

marketshare = 0.0403 - 0.000760 Hannah + 0.000262 Petra + 0.000347 Efacec1 + 0.000102 Efacec2

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## Regression Analysis: marketshare versus Hannah, Petra, Efacec1

### Analysis of Variance

| Source      | DF  | Adj SS   | Adj MS   | F-Value | P-Value |
|-------------|-----|----------|----------|---------|---------|
| Regression  | 3   | 0.012002 | 0.004001 | 40.29   | 0.000   |
| Hannah      | 1   | 0.009160 | 0.009160 | 92.24   | 0.000   |
| Petra       | 1   | 0.000979 | 0.000979 | 9.86    | 0.002   |
| Efacec1     | 1   | 0.003434 | 0.003434 | 34.58   | 0.000   |
| Error       | 109 | 0.010824 | 0.000099 |         |         |
| Lack-of-Fit | 47  | 0.006282 | 0.000134 | 1.82    | 0.013   |
| Pure Error  | 62  | 0.004542 | 0.000073 |         |         |
| Total       | 112 | 0.022826 |          |         |         |

### Model Summary

| S         | R-sq   | R-sq(adj) | R-sq(pred) |
|-----------|--------|-----------|------------|
| 0.0099651 | 52.58% | 51.27%    | 48.33%     |

### Coefficients

| Term     | Coef      | SE Coef  | T-Value | P-Value | VIF  |
|----------|-----------|----------|---------|---------|------|
| Constant | 0.0401    | 0.0140   | 2.85    | 0.005   |      |
| Hannah   | -0.000764 | 0.000080 | -9.60   | 0.000   | 1.33 |
| Petra    | 0.000263  | 0.000084 | 3.14    | 0.002   | 1.66 |
| Efacec1  | 0.000460  | 0.000078 | 5.88    | 0.000   | 1.45 |

### Regression Equation

marketshare = 0.0401 - 0.000764 Hannah + 0.000263 Petra + 0.000460 Efacec1

Leave Efacec2 out of the regression

Which prices are significant with 5% level of significance?

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What's Happening?

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## Fit Regression v.s. Make a Margarita

### How to make a Margarita?

- **Ingredient Selection:** Both require choosing the right inputs—predictors for a model and ingredients (tequila, lime juice, etc) for a margarita.
- **Proportion and Balance:** Achieving the right mix in both processes involves balancing ingredient ratios in a margarita and fitting coefficients in regression.
- **Evaluation:** After preparation, both require assessing quality—tasting the margarita or analyzing model diagnostics for fit and accuracy.

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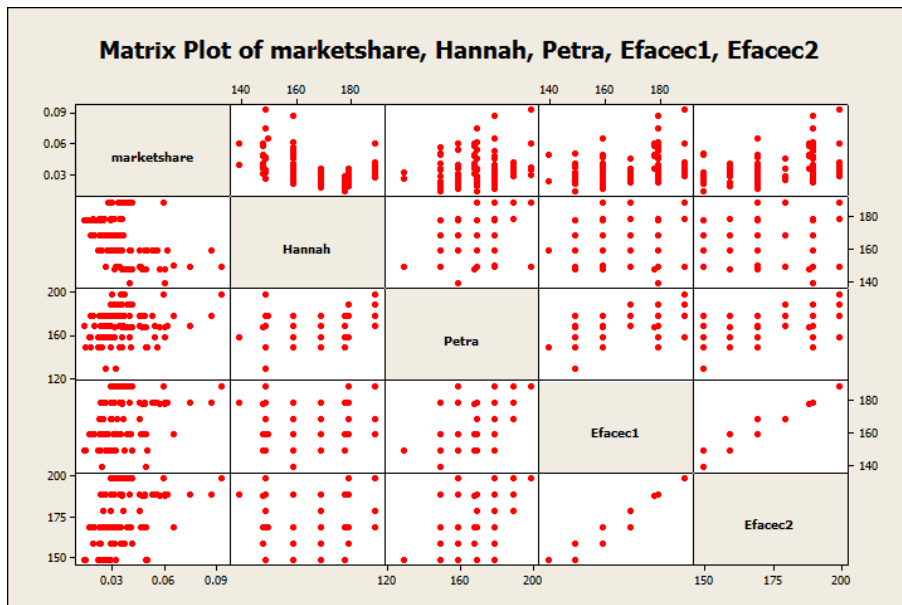
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## Multicollinearity

Independent variables are highly correlated!

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**Correlations: marketshare, Hannah, Petra, Efacec1, Efacec2**

|         | marketshare     | Hannah         | Petra          | Efacec1        |
|---------|-----------------|----------------|----------------|----------------|
| Hannah  | -0.433<br>0.000 |                |                |                |
| Petra   | 0.170<br>0.073  | 0.484<br>0.000 |                |                |
| Efacec1 | 0.352<br>0.000  | 0.359<br>0.000 | 0.549<br>0.000 |                |
| Efacec2 | 0.369<br>0.000  | 0.323<br>0.000 | 0.534<br>0.000 | 0.979<br>0.000 |

Cell Contents: Pearson correlation  
P-Value

## Detecting Multicollinearity

- Examine correlation matrix.
- Examine **variance inflation factor (VIF)**
  - If  $VIF > 5$ , multicollinearity exists.

$$VIF(X_j) = \frac{1}{1 - R_j^2}$$

where  $R_j^2$  is the coefficient of multiple determination of explanatory variable  $X_j$  **when we regress  $X_j$  with all other  $X$  variables.**

## Testing joint significance

### Partial F-test for Efacec1, Efacec2

- Our original question: is Efacec a significant competitor?  
**That is : Do Efacec1, Efacec2 **together** significantly influence Hannah's marketshare?**
- Hannah, Petra prices already explain about Hannah's marketshare. *How much more* Efacec1 and Efacec2 **together** can explain Hannah's marketshare?
- Compare the **Explained Variations** of two model:
  - Model without Efacec1 and Efacec2  
**marketshare on Hannah, Petra**
  - Full Model  
**marketshare on Hannah, Petra, Efacec1, Efacec2**



## Regression Analysis: marketshare versus Hannah, Petra

### Analysis of Variance

| Source      | DF  | Adj SS   | Adj MS   | F-Value | P-Value |
|-------------|-----|----------|----------|---------|---------|
| Regression  | 2   | 0.008568 | 0.004284 | 33.05   | 0.000   |
| Hannah      | 1   | 0.007912 | 0.007912 | 61.04   | 0.000   |
| Petra       | 1   | 0.004290 | 0.004290 | 33.09   | 0.000   |
| Error       | 110 | 0.014258 | 0.000130 |         |         |
| Lack-of-Fit | 24  | 0.006264 | 0.000261 | 2.81    | 0.000   |
| Pure Error  | 86  | 0.007994 | 0.000093 |         |         |
| Total       | 112 | 0.022826 |          |         |         |

### Model Summary

| S         | R-sq   | R-sq(adj) | R-sq(pred) |
|-----------|--------|-----------|------------|
| 0.0113849 | 37.54% | 36.40%    | 33.46%     |

### Coefficients

| Term     | Coef      | SE Coef  | T-Value | P-Value | VIF  |
|----------|-----------|----------|---------|---------|------|
| Constant | 0.0688    | 0.0150   | 4.58    | 0.000   |      |
| Hannah   | -0.000704 | 0.000090 | -7.81   | 0.000   | 1.31 |
| Petra    | 0.000490  | 0.000085 | 5.75    | 0.000   | 1.31 |

### Regression Equation

marketshare = 0.0688 - 0.000704 Hannah + 0.000490 Petra

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## Regression Analysis: marketshare versus Hannah, Petra, Efacec1, Efacec2

### Analysis of Variance

| Source      | DF  | Adj SS   | Adj MS   | F-Value | P-Value |
|-------------|-----|----------|----------|---------|---------|
| Regression  | 4   | 0.012014 | 0.003003 | 30.00   | 0.000   |
| Hannah      | 1   | 0.008826 | 0.008826 | 88.16   | 0.000   |
| Petra       | 1   | 0.000969 | 0.000969 | 9.68    | 0.002   |
| Efacec1     | 1   | 0.000110 | 0.000110 | 1.10    | 0.297   |
| Efacec2     | 1   | 0.000012 | 0.000012 | 0.12    | 0.728   |
| Error       | 108 | 0.010812 | 0.000100 |         |         |
| Lack-of-Fit | 49  | 0.006692 | 0.000137 | 1.96    | 0.007   |
| Pure Error  | 59  | 0.004120 | 0.000070 |         |         |
| Total       | 112 | 0.022826 |          |         |         |

### Model Summary

| S         | R-sq   | R-sq(adj) | R-sq(pred) |
|-----------|--------|-----------|------------|
| 0.0100055 | 52.63% | 50.88%    | 47.57%     |

### Coefficients

| Term     | Coef      | SE Coef  | T-Value | P-Value | VIF   |
|----------|-----------|----------|---------|---------|-------|
| Constant | 0.0403    | 0.0141   | 2.85    | 0.005   |       |
| Hannah   | -0.000760 | 0.000081 | -9.39   | 0.000   | 1.36  |
| Petra    | 0.000262  | 0.000084 | 3.11    | 0.002   | 1.66  |
| Efacec1  | 0.000347  | 0.000332 | 1.05    | 0.297   | 25.97 |
| Efacec2  | 0.000102  | 0.000294 | 0.35    | 0.728   | 25.15 |

### Regression Equation

marketshare = 0.0403 - 0.000760 Hannah + 0.000262 Petra + 0.000347 Efacec1 + 0.000102 Efacec2

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### REDUCE MODEL: Regression Analysis: marketshare versus Hannah, Petra

| Source         | DF  | SS        | MS        | F     | P     |
|----------------|-----|-----------|-----------|-------|-------|
| Regression     | 2   | 0.0085679 | 0.0042839 | 33.05 | 0.000 |
| Residual Error | 110 | 0.0142579 | 0.0001296 |       |       |
| Total          | 112 | 0.0228257 |           |       |       |

### FULL MODEL: Regression Analysis: marketshare versus Hannah, Petra, Efacec1, Efacec2

| Source         | DF  | SS        | MS        | F     | P     |
|----------------|-----|-----------|-----------|-------|-------|
| Regression     | 4   | 0.0120140 | 0.0030035 | 30.00 | 0.000 |
| Residual Error | 108 | 0.0108118 | 0.0001001 |       |       |
| Total          | 112 | 0.0228257 |           |       |       |

$$\text{partial F-test} = \frac{\frac{\text{SSR}(\text{Full}) - \text{SSR}(\text{Reduced})}{\# \text{variables}(\text{Full}) - \# \text{variable}(\text{Reduced})}}{\frac{\text{SSE}(\text{Full})}{n - \# \text{variables}(\text{Full}) - 1}} =$$

Critical value =  $F(\alpha, \# \text{variable}(\text{full}) - \# \text{variable}(\text{reduced}), n - \# \text{variable}(\text{full}) - 1)$

example  $\alpha = 5\%$

## Conclusion

- Even though individually Efacec1, Efacec2 are not significant, together they are!
- Thus, Efacec is a significant competitor

## Models and Fits

$$\text{marketshare} = \beta_0 + \beta_1 \times \text{Hannah} + \beta_2 \times \text{Petra} + \beta_3 \times \text{Efacec1} + \beta_4 \times \text{Efacec2} + \varepsilon$$

$$S = 0.0100055 \quad R\text{-Sq} = 52.6\% \quad R\text{-Sq}(\text{adj}) = 50.9\%$$

$$\text{marketshare} = \beta_0 + \beta_1 \times \text{Hannah} + \beta_2 \times \text{Petra} + \beta_3 \times \text{Efacec1} + \varepsilon$$

$$S = 0.00996506 \quad R\text{-Sq} = 52.6\% \quad R\text{-Sq}(\text{adj}) = 51.3\%$$

$$\text{marketshare} = \beta_0 + \beta_1 \times \text{Hannah} + \beta_2 \times \text{Petra} + \beta_4 \times \text{Efacec2} + \varepsilon$$

$$S = 0.0100099 \quad R\text{-Sq} = 52.2\% \quad R\text{-Sq}(\text{adj}) = 50.8\%$$

$$\text{marketshare} = \beta_0 + \beta_1 \times \text{Hannah} + \beta_2 \times \text{Petra} + \varepsilon$$

$$S = 0.0113849 \quad R\text{-Sq} = 37.5\% \quad R\text{-Sq}(\text{adj}) = 36.4\%$$

## Prediction?

- What would be the market share if
  - Hannah price: 149, Petra Price: 169,
  - Efacec1 price: 160 and Efacec2 price: 169
- Choose model:

$$\text{marketshare} = \beta_0 + \beta_1 \times \text{Hannah} + \beta_2 \times \text{Petra} + \beta_3 \times \text{Efacec1} + \varepsilon$$

Can also specify columns where data values for each variable are stored

### Predicted Values for New Observations

| New Obs | Fit      | SE Fit   | 95% CI               | 95% PI               |
|---------|----------|----------|----------------------|----------------------|
| 1       | 0.044250 | 0.001694 | (0.040893, 0.047608) | (0.024217, 0.064284) |

### Values of Predictors for New Observations

| New Obs | Hannah | Petra | Efacec1 |
|---------|--------|-------|---------|
| 1       | 149    | 169   | 160     |

How to handle categorical predictors?

| store ID | endcap | sales2012 in \$K | population 3mi | income 3mi | yelp_rating |
|----------|--------|------------------|----------------|------------|-------------|
| 1        | 1      | 1651             | 15432          | 47644      | 1.5         |
| 2        | 1      | 3517             | 51425          | 31273      | 4           |
| 3        | 1      | 3431             | 54496          | 67161      | 4           |
| 4        | 1      | 1328             | 63234          | 47911      | 1           |
| 5        | 1      | 4677             | 64222          | 49429      | 5           |
| 6        | 1      | 4699             | 76547          | 91781      | 5           |
| 7        | 1      | 3704             | 81091          | 45824      | 4           |
| 8        | 1      | 3907             | 81495          | 61567      | 4.5         |
| 9        | 1      | 2441             | 99660          | 54657      | 2.5         |
| 10       | 1      | 2550             | 102122         | 53474      | 2.5         |
| 11       | 1      | 3887             | 111246         | 61989      | 4.5         |
| 12       | 1      | 5112             | 116345         | 56149      | 5           |
| 13       | 1      | 4677             | 118329         | 58673      | 5           |
| 14       | 1      | 2933             | 120915         | 26588      | 3           |
| 15       | 1      | 4733             | 133281         | 68709      | 5           |
| 16       | 1      | 4607             | 135325         | 48502      | 5           |
| 17       | 1      | 3375             | 150982         | 33792      | 5           |
| 18       | 1      | 1764             | 182381         | 35875      | 1.5         |
| 19       | 1      | 4360             | 387887         | 39480      | 5           |
| 20       | 1      | 3444             | 389167         | 38821      | 4           |
| 21       | 1      | 2285             | 396904         | 38957      | 2           |
| 22       | 1      | 2734             | 459473         | 47320      | 3           |
| 23       | 1      | 2496             | 527877         | 51200      | 2           |
| 24       | 0      | 2631             | 27486          | 110148     | 4           |
| 25       | 0      | 2165             | 35436          | 64938      | 3           |
| 26       | 0      | 2869             | 36788          | 78608      | 5           |
| 27       | 0      | 1926             | 65415          | 50281      | 2.5         |
| 28       | 0      | 3291             | 66395          | 82936      | 5           |
| 29       | 0      | 2302             | 80210          | 58287      | 3           |
| 30       | 0      | 1913             | 81438          | 65130      | 2.5         |
| 31       | 0      | 1681             | 105833         | 88587      | 2           |
| 32       | 0      | 1272             | 109139         | 106508     | 1           |
| 33       | 0      | 2128             | 110247         | 73393      | 3           |
| 34       | 0      | 2362             | 115501         | 66667      | 3.5         |
| 35       | 0      | 2069             | 121059         | 56754      | 3           |
| 36       | 0      | 1059             | 125624         | 60894      | 1           |
| 37       | 0      | 1691             | 127477         | 73027      | 2           |
| 38       | 0      | 1399             | 156559         | 63554      | 1           |
| 39       | 0      | 1695             | 163038         | 36416      | 2           |
| 40       | 0      | 1502             | 171086         | 63085      | 1.5         |
| 41       | 0      | 2148             | 192373         | 44851      | 3           |
| 42       | 0      | 2033             | 193961         | 69780      | 2.5         |
| 43       | 0      | 1460             | 250989         | 60684      | 1.5         |
| 44       | 0      | 1690             | 380860         | 62821      | 2           |
| 45       | 0      | 2186             | 386399         | 44625      | 3           |

## Auto Parts Case Study

## Dummy Variable for Categorical Predictors

- Endcap: 1 and 0 (Auto parts data)
- **Categorical predictors with K levels -> create K-1 dummy variables.**
- E.g. Education level = {High school, Undergraduate, Graduate}. create dummy variables:
  - X1 = Is\_High\_School? = {0,1}
  - X2 = Is\_Undergraduate? = {0,1}
- In practice, multiplying it with other independent variables might increase prediction power

## Summary

- Adjusted R-square
- Different tests
  - Overall test;
  - Individual test;
  - partial F-test
- Multicollinearity
  - VIF
- Dummy variable